

Linear Algebra — Concept Map

how we built from idea to idea

study notes

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1 The mind map



2 How we built: concept → concept

1. **Vector space** — the foundation. A set with addition and scalar multiplication (closure, zero vector, axioms). A vector is any element; in $\mathbb{R}^n$ it's a stack of numbers.
2. **Representation → dimension.** $\mathbb{R}^n$ has n directions, P_n has $n + 1$, P has $\aleph_0$, C is uncountable. Dimension = number of independent directions (basis size), not vector length.

3. **Free vs fixed.** Displacement vs position reading; the math runs on displacements (a basis, never an origin). This is why $[3, 5]$ needs no reference point as a displacement.
4. **Norm.** Add “size” to a vector space (3 axioms: positive definiteness, homogeneity, triangle inequality). Different norms = different geometry (unit spheres: circle, diamond, square, ellipse).
5. **Metric / distance.** Every norm induces a metric $d(x, y) = \|x - y\|$; “ $\|x\|$ = distance to the origin” is the position-reading corollary, not the definition.
6. **Triangle inequality.** Its engine is the projection split: $v = v_{\parallel} + v_{\perp}$, and $\|v_{\parallel}\| \leq \|v\|$ — a projection is never longer than the vector it came from.
7. **Cauchy–Schwarz.** That engine in dot-product clothes: $|\langle u, v \rangle| \leq \|u\| \|v\|$. Same fact, two languages.
8. **Inner product.** Add “alignment” between two vectors ($\langle x, y \rangle = x^T y$). It induces the norm $\|x\| = \sqrt{\langle x, x \rangle}$ and brings angles, orthogonality, and projections.
9. **Parallelogram law / polarization.** The bridge back: a norm comes from an inner product iff $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$ (Jordan–von Neumann), and polarization recovers the inner product from the norm — the norm was the inner product in disguise.
10. **Weighted inner products.** Any symmetric positive-definite M works ($\langle x, y \rangle_M = x^T M y$); the M norm is one of them. ℓ_2 is the flagship ($M = I$), the only rotation-invariant norm.

3 The ladder of structure

Every step adds structure; the richest sits at the top.

$$\begin{array}{ccccc}
 & \text{inner product space} & \subset & \text{normed space} & \subset \text{metric space} \\
 & (\text{sizes} + \text{angles}) & & (\text{sizes}) & (\text{just distances}) \\
 \text{vector space is the base: set} & + \text{addition} & + \text{scaling} & &
 \end{array}$$

- **vector space** $\rightarrow$ **normed space**: add $\|\cdot\|$ (3 axioms).
- **normed space** $\rightarrow$ **inner product space**: add $\langle \cdot, \cdot \rangle$, which induces the norm.
- **normed space** $\rightarrow$ **metric space**: every norm gives $d(x, y) = \|x - y\|$ (the triangle inequality is the norm’s own axiom).
- Each step up is one-directional: inner product $\Rightarrow$ norm $\Rightarrow$ metric, but not back (except when the parallelogram law holds).